

Fresh Enough to Decide: Age of Intelligence for Cost-Aware Model Synchronization in Dynamic IoT Systems

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Abstract

Resource-constrained IoT devices cache learned models for real-time decisions, but frequent updates incur substantial bandwidth, energy, and service overhead. Because decision-relevant model differences need not grow with elapsed time, we define *Age of Intelligence* (AoInt) as a model-version-relative, client-specific, and action utility distortion, and formulate synchronization as a *tradeoff between stale-decision regret and refresh cost*. Furthermore, we develop Kullback-Leibler-aware AoInt Synchronization (KL-AoIntSYNC), which accumulates output-distribution changes using client-specific unlabeled probes, whereas Parameter-aware AoInt Synchronization (PARAM-AoIntSYNC) uses a server-broadcast scalar parameter change. Then, we establish pathwise guarantees with fixed client distributions, action sets, and utilities. The numerical results show that calibrated PARAM-AoIntSYNC reduces joint synchronization cost by 49% relative to a tuned periodic policy while performing comparably to KL-AoIntSYNC.

CCS Concepts

• **Networks** → **Network performance modeling**; • **Computer systems organization** → **Internet of things**; • **Computing methodologies** → *Machine learning approaches*.

Keywords

Age of Intelligence, Edge Intelligence, IoT, Model Freshness, Model Synchronization Trade-Offs, Nonstationary Learning.

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1 Introduction

The Internet of Things (IoT) increasingly relies on device-side machine-learning models for latency-sensitive prediction and control in applications such as industrial automation, transportation, health monitoring, spatial computing, and smart infrastructure [2,

8, 14, 22]. Executing inference near the data source can reduce response latency and raw-data exposure, and can sustain operation when cloud connectivity is intermittent. But, the environments observed by IoT devices are often dynamic: sensing conditions, user behavior, equipment state, and task requirements may change over time. A model suitable when installed can consequently become misaligned with its present operating conditions.

Resource constraints make this problem especially challenging. Many devices can execute a compact inference model but cannot continuously retrain it, retain numerous versions, or repeatedly download large updates. A common architecture therefore updates models at an edge server and lets each device perform inference with a locally cached copy. This separation creates a downstream deployment decision distinct from model training: when should a device replace its cached model with the latest server version? Every refresh consumes bandwidth and energy, occupies memory, and may interrupt service while the new model is validated and installed. Conversely, postponing a consequential update can degrade the decisions made with stale intelligence. Communication-efficient distributed-training methods, including federated learning, adaptive aggregation, event-triggered SGD, and gossip-based learning [9, 13, 18, 21], reduce the cost of producing a model but do not determine whether an already trained upstream model is worth deploying to a particular device.

A natural baseline is to refresh periodically or according to the elapsed time since the previous update. Age of Information (AoI) formalizes this chronological notion of freshness [11, 17, 20]. Yet the operational value of a model need not decay monotonically with time. During stable operation, an old model may remain prediction- or decision-equivalent to the latest version; after an abrupt change, even a recently installed model may become obsolete. This observation has motivated richer freshness metrics that incorporate correctness, estimation error, goals, or value [3, 7, 10, 12]. For learning services, Age of Model measures inference-performance degradation [5], while Age of Context measures the relevance of the samples supporting an edge-intelligence service [19]. These approaches establish that task impact matters, but they do not directly supply the low-rate signal needed for cost-aware model deployment.

In particular, a constrained device often holds only its cached model and cannot download and execute every candidate merely to assess its value. Such an assessment may require storing two model versions, evaluating representative inputs, and validating de-

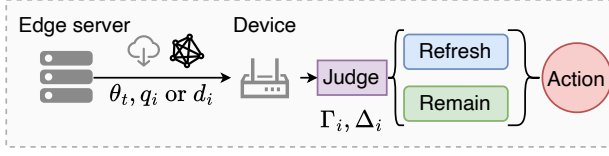


Figure 1: Overview of the proposed AoInt-Aware synchronization framework.

cisions before installation. Moreover, the labels or rewards needed to measure decision performance may be delayed, sparse, or unavailable online. Heterogeneous contexts, actions, utilities, and refresh costs also mean that a single server-side accuracy score may not characterize the value of an update for every device. An effective synchronization mechanism must detect consequential model evolution before deployment, operate without online supervision, accumulate gradual changes, and expose a tunable tradeoff between decision degradation and refresh cost.

We cast this synchronization task as a nonstationary decision problem with an explicit refresh cost, drawing on variation budgets, dynamic regret, and switching-cost learning [1, 4, 6, 15, 16]. We introduce *Age of Intelligence* (AoInt), the expected utility loss caused by acting with a device's cached model rather than the current server model. Unlike elapsed time or raw parameter distance, AoInt directly measures the decision consequence of staleness and therefore distinguishes model motion from model changes that actually affect downstream actions.

Our contributions are threefold. First, we formulate cost-aware model synchronization using AoInt, a client-specific and action-utility distortion between the current server model and a device's cached model. We then derive a Kullback–Leibler (KL) output-space certificate satisfying $\text{AoInt}_{i,t} = r_{i,t}(\theta_{i,t}) \leq \kappa \widehat{C}_{i,t}^{\text{KL}}$ with high probability, where $r_{i,t}$ denotes the instant regret of device i under cached model parameter $\theta_{i,t}$ and $\widehat{C}_{i,t}^{\text{KL}}$ denote the approximate KL divergence between cached model and server model. Second, we develop KL-aware Age-of-Intelligence Synchronization (KL-AoIntSYNC), which accumulates predictive-distribution changes evaluated on client-specific unlabeled probes, and its lightweight Parameter-aware Age-of-Intelligence Synchronization (PARAM-AoIntSYNC), which uses a single scalar parameter-change signal broadcast by the server. For both methods, we establish pathwise bounds on stale-model regret, refresh count, and their joint cost, as well as sublinear regret when cumulative server-model variation is sublinear. Third, simulations with heterogeneous clients and server-model paths containing both gradual drift and abrupt jumps show that PARAM-AoIntSYNC reduces joint cost by 6.15% and regret by 9.22% relative to a Instant-Parameters policy, while achieving performance comparable to KL-AoIntSYNC. The proposed framework and its key components are summarized in Fig. 1.

2 Problem Formulation and Age of Intelligence

We consider an IoT system with N heterogeneous devices $\mathcal{N} = \{1, \dots, N\}$ over a horizon $\mathcal{T} = \{1, \dots, T\}$. At each round $t \in \mathcal{T}$, the server maintains its latest model $\theta_t \in \Theta$, while device $i \in \mathcal{N}$ operates with a cached model $\theta_{i,t} \in \{\theta_\tau\}_{\tau \in \{1, \dots, t\}}$. Before observing or acting, the device either retains its cached model or syn-

chronizes with θ_t at cost c_i . Specifically, device i makes a binary synchronization decision $z_{i,t} \in \{0, 1\}$ and updates its model as $\theta_{i,t} = z_{i,t}\theta_t + (1 - z_{i,t})\theta_\tau$ ($\tau \in [1, t)$). Here, $z_{i,t} = 0$ retains the cached model, whereas $z_{i,t} = 1$ installs the current server model θ_t before the round- t action and incurs a normalized refresh cost $c_i \in \mathbb{R}_+$. The total number of model refreshes is $M_i(T) := \sum_{t=1}^T z_{i,t}$. The synchronization sequence is generated by a policy $\pi : \Theta \times \Theta \mapsto \{0, 1\}$, and Π_c denotes the class of causal policies that use only information available up to round t .

Next, a latent context $s_{i,t} \in \mathcal{S}$ is drawn from μ_i . Due to space limitations, we consider a fixed context distribution and leave extensions to time-varying distributions to future work. Conditional on this context, $\theta_{i,t}$ induces a predictive distribution $\mathbb{P}_{\theta_{i,t}}(y_{i,t} | s_{i,t})$ over the outcome space $\mathcal{Y}$. Based on the model output and its utility function $u_i : \mathcal{Y} \times \mathcal{A} \mapsto [0, 1]$, device i selects an action $a_{i,t} \in \mathcal{A}$. We impose the following assumption before introducing the performance metrics.

ASSUMPTION 2.1. *For each device $i \in \mathcal{N}$, the distribution μ_i , $\mathcal{S}$ with $|\mathcal{S}| = S < \infty$, $\mathcal{A}$ with $|\mathcal{A}| = A < \infty$, $\mathcal{Y}$ with $|\mathcal{Y}| = Y < \infty$, and $c_i \in \mathbb{R}_+$ remain fixed over $\mathcal{T}$, where $|\cdot|$ denotes cardinality.*

Definition 2.1 (Decision Utility, Regret, and AoInt). For each device $i \in \mathcal{N}$, let $h_i : \mathcal{A}_i \times \mathcal{S} \rightarrow [0, 1]$ be a known utility function. The optimal expected cumulative reward under model θ is $V_i^\theta = \sup_{a_{i,t} \in \mathcal{A}_i} \mathbb{E}_{s_{i,t} \sim \mu_i, y_{i,t} \sim \mathbb{P}_\theta(\cdot | s_{i,t})} [\sum_{t \in \mathcal{T}} h_i(a_{i,t}, y_{i,t})]$, and the corresponding oracle value is $V_i^* = \sup_{a_{i,t} \in \mathcal{A}_i} \mathbb{E}_{s_{i,t} \sim \mu_i} [\sum_{t \in \mathcal{T}} h_i(a_{i,t}, s_{i,t})]$. The regret of device i can therefore be decomposed as

$$\text{Reg}_i = V_i^* - V_i^{\theta_{i,t}} = V_i^* - V_i^{\theta_t} + (V_i^{\theta_t} - V_i^{\theta_{i,t}}). \quad (1)$$

This decomposition separates the oracle gap of the current server model from the additional gap associated with the cached model. Accordingly, we define AoInt as

$$\text{AoInt}_{i,t} = \sup_{a \in \mathcal{A}_i} \mathbb{E}_{\theta_t} h_i(a, y_{i,t}) - \sup_{a \in \mathcal{A}_i} \mathbb{E}_{\theta_{i,t}} h_i(a, y_{i,t}), \quad (2)$$

where $\mathbb{E}_\theta$ denotes $\mathbb{E}_{y_{i,t} \sim \mathbb{P}_\theta}$. The cumulative AoInt bounds the corresponding value gap: $\sum_{t \in \mathcal{T}} \text{AoInt}_{i,t} \geq V_i^{\theta_t} - V_i^{\theta_{i,t}}$.

To focus on evaluating the synchronization process, we impose the following assumption on the current server model.

ASSUMPTION 2.2. *For each device i , the current server model θ_t is ϵ -optimal that satisfies $V_i^* - V_i^{\theta_t} \leq \epsilon, \epsilon > 0$, so that $\text{Reg}_i \leq \epsilon + (V_i^{\theta_t} - V_i^{\theta_{i,t}}) \leq \epsilon + \sum_{t \in \mathcal{T}} \text{AoInt}_{i,t}$.*

AoInt measures a difference in expected action utility rather than an observed realized reward loss. It may be positive even when the maximizing action remains unchanged. Moreover, unlike chronological AoI, it need not increase monotonically between refreshes, because the server model may move toward or away from the cached version. The AoInt component measures deployment inconsistency relative to this reference, without implying that every newer model is necessarily better for the client's true environment. In the following, we write $r_{i,t} = \text{AoInt}_{i,t}$.

Definition 2.2 (Joint Cost). The joint cost incurred by π over the horizon combines stale-model regret and synchronization cost, where $r_{i,t}^\pi = r_{i,t}(\pi(\theta_t, \theta_{t-1})\theta_t + (1 - \pi(\theta_t, \theta_{t-1}))\theta_{i,t})$:

$$J_T^\pi := \sum_{i=1}^N \sum_{t=1}^T [r_{i,t}^\pi + c_i \pi(\theta_t, \theta_{t-1})]. \quad (3)$$

Definition 2.3 (Output-KL Certificate). For any models θ and θ' , we define the KL certificate as

$$C_i^{\text{KL}}(\theta, \theta') := \sqrt{\frac{1}{2} \mathbb{E}_{s \sim \mu_i} D_{\text{KL}}(p_\theta(\cdot | s) \| p_{\theta'}(\cdot | s))}. \quad (4)$$

Since the online context $s_{i,t}$ and its distribution μ_i are unknown, we estimate it by given probes $\{s_i^{(m)}\}_{m=1}^{M_i}$, and estimate it by

$$\widehat{C}_i^{\text{KL}}(\theta, \theta') := \sqrt{\frac{1}{2M_i} \sum_{m=1}^{M_i} D_{\text{KL}}(p_\theta(\cdot | s_i^{(m)}) \| p_{\theta'}(\cdot | s_i^{(m)})}. \quad (5)$$

Henceforth, we write $C_{i,t}^{\text{KL}} := C_i^{\text{KL}}(\theta_t, \theta_{i,t})$ and $\widehat{C}_{i,t}^{\text{KL}} := \widehat{C}_i^{\text{KL}}(\theta_t, \theta_{i,t})$.

Next, we bound the magnitude of AoInt using the predictive-distribution discrepancy between the current and cached models.

PROPOSITION 2.1 (EMPIRICAL KL CONTROL OF AOINT). *Under Definitions 2.1 and 2.3, fix $(\theta_t, \theta_{i,t})$ independently of M_i i.i.d. probes from μ_i . Let $X(s) := D_{\text{KL}}(p_{\theta_t}(\cdot | s) \| p_{\theta_{i,t}}(\cdot | s)) \in [0, B_i]$ almost surely and $D := \mathbb{E}_{\mu_i} X > 0$. For $\kappa > 1$ and $\delta \in (0, 1)$, if $M_i \geq B_i^2 \log(1/\delta) / [2D^2(1 - \kappa^{-2})^2]$, then, with probability at least $1 - \delta$, $|\text{AoInt}_{i,t}| = |r_{i,t}(\theta_{i,t})| \leq C_{i,t}^{\text{KL}} < \kappa \widehat{C}_{i,t}^{\text{KL}}$.*

PROOF. Bounded utilities, Pinsker's inequality, $|\max f - \max g| \leq \max |f - g|$, and Jensen's inequality give $|\text{AoInt}_{i,t}| \leq \mathbb{E} \sqrt{X/2} \leq \sqrt{D/2} = C_{i,t}^{\text{KL}}$. Hoeffding's inequality gives $\Pr(C_{i,t}^{\text{KL}} \geq \kappa \widehat{C}_{i,t}^{\text{KL}}) \leq e^{-2M_i D^2 (1 - \kappa^{-2})^2 / B_i^2} \leq \delta$, proving the claim. $\square$

3 Algorithm Design

We now develop synchronization policies π that can be evaluated before a device installs the current server model. Proposition 2.1 shows that the empirical KL certificate controls AoInt, and hence stale-model regret, with high probability. In particular, for a fixed model pair satisfying the probe requirement of Proposition 2.1, $r_{i,t}(\theta_{i,t}) \leq C_{i,t}^{\text{KL}} < \kappa \widehat{C}_{i,t}^{\text{KL}}$ with the corresponding high probability. This relation provides the basis for the empirical synchronization certificates developed below.

Direct evaluation of the endpoint discrepancy, however, requires comparing the current server model with the device's cached version. We therefore begin with KL-AOINTSYNC, which accumulates output-space changes between consecutive server models, and then introduce PARAM-AOINTSYNC, which replaces client-specific output comparisons with $\|\theta_t - \theta_{t-1}\|_F$. Both mechanisms use the same timing. At $t = 1$, device i is provisioned with θ_1 , initializes its accumulator to zero, and acts with $\theta_{i,1} = \theta_1$. For each $t \geq 2$, it inherits the previous cache, receives the relevant beacon, updates the accumulator tentatively, tests the threshold, and, if triggered, installs θ_t and resets the accumulator $Q_i, P_i = 0$ before taking its round- t action. Otherwise, it retains the inherited cache $\theta_{i,t}$ and the updated accumulator. An overview is provided in Fig. 2.

3.1 Kullback–Leibler-Aware Age-of-Intelligence Synchronization

The key design parameter governing Algorithm 1 is its synchronization threshold. Following Proposition 2.1, we define the KL-based change between consecutive server models as $q_{i,t} := \kappa \widehat{C}_{i,t}^{\text{KL}}$. For each adjacent model pair, Proposition 2.1 implies that $C_i^{\text{KL}}(\theta_t, \theta_{t-1}) < q_{i,t}$ with high probability whenever its probe requirement is satisfied.

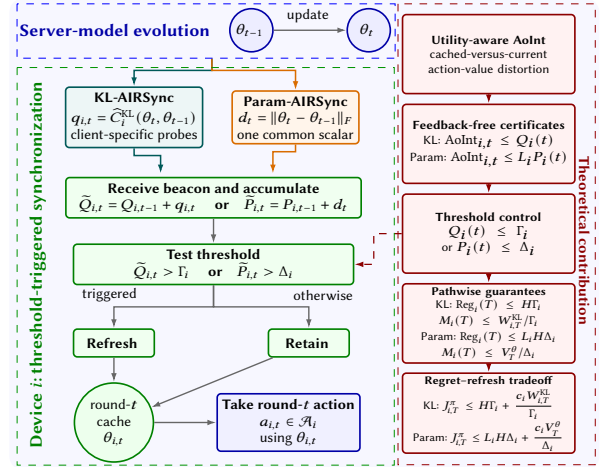


Figure 2: An illustration of Algorithm 1 in an edge-AI system.

To obtain a horizon-wide guarantee, let $H := T - 1$ and apply Proposition 2.1 to every adjacent model pair with failure probability δ/H . Accordingly, it is sufficient that the probe requirement in Proposition 2.1 be satisfied with δ replaced by δ/H . In particular, for each adjacent model pair, the corresponding requirement becomes $M_i \geq B_i^2 \log(H/\delta) / [2D^2(1 - \kappa^{-2})^2]$. A union bound then ensures that all adjacent empirical certificates simultaneously dominate their population counterparts with probability at least $1 - \delta$.

In practice, each client i provides a fixed unlabeled probe set $\mathcal{S}$ once. The server evaluates each pair of adjacent models on this set and unicasts the resulting scalar $q_{i,t}$. The signal is client-specific through the client's context distribution, yet it requires neither labels nor observed utilities. To prevent a sequence of individually small changes from going undetected, device i accumulates the adjacent certificates since its most recent synchronization: $Q_i(t) := \sum_{\ell=\tau_{i,t}+1}^t q_{i,\ell}$. Because each $q_{i,\ell}$ is an empirical high-probability upper certificate rather than a deterministic population certificate, the resulting control of AoInt is itself a high-probability statement. The following lemma makes this relation explicit.

LEMMA 3.1 (CUMULATIVE KL CERTIFICATE). *Under a fixed context distribution μ_i and action set $\mathcal{A}_i$, suppose that the probe requirement of Proposition 2.1 holds for each adjacent model pair after the most recent synchronization. Define $\mathcal{E}_{i,t} := \bigcap_{\ell=\tau_{i,t}+1}^t \{C_i^{\text{KL}}(\theta_\ell, \theta_{\ell-1}) < q_{i,\ell}\}$. Then, on $\mathcal{E}_{i,t}$, $\text{AoInt}_{i,t} = r_{i,t}(\theta_{i,t}) < Q_i(t)$. Moreover, if each adjacent certificate has failure probability at most δ/H , then*

$$\Pr(\mathcal{E}_{i,t}) \geq 1 - (t - \tau_{i,t})\delta/H \geq 1 - \delta. \quad (6)$$

PROOF. Fix t and write $\tau := \tau_{i,t}$. The difference between the current and cached expected utilities can be decomposed along the server-model path from θ_τ to θ_t . Applying the triangle inequality gives $\text{AoInt}_{i,t} = r_{i,t}(\theta_{i,t}) \leq \sum_{\ell=\tau+1}^t C_i^{\text{KL}}(\theta_\ell, \theta_{\ell-1})$. On the event $\mathcal{E}_{i,t}$, Proposition 2.1 gives $C_i^{\text{KL}}(\theta_\ell, \theta_{\ell-1}) < q_{i,\ell} = \kappa \widehat{C}_{i,\ell}^{\text{KL}}$ for every $\ell = \tau + 1, \dots, t$. Therefore, $\text{AoInt}_{i,t} = r_{i,t}(\theta_{i,t}) \leq \sum_{\ell=\tau+1}^t C_i^{\text{KL}}(\theta_\ell, \theta_{\ell-1}) < Q_i(t)$. Finally, each adjacent-pair certificate fails with probability at most δ/H . Applying the union bound over $t - \tau_{i,t} \leq H$ adjacent model pairs gives $\Pr(\mathcal{E}_{i,t}) \geq 1 - (t - \tau_{i,t})\delta/H \geq 1 - \delta$. $\square$

Algorithm 1 Certificate-Based AIRSync at device i

Require: Initial model θ_1 and threshold Γ_i for KL-AOINTSYNC or Δ_i for PARAM-AOINTSYNC

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1:  $z_{i,1} \leftarrow 0$ ,  $\theta_{i,1} \leftarrow \theta_1$ , and  $\tau_{i,1} \leftarrow 1$ 
2:  $Q_i \leftarrow 0$  for KL-AOINTSYNC or  $P_i \leftarrow 0$  for PARAM-AOINTSYNC
3: Observe  $s_{i,1}$  and select  $a_{i,1}$  using  $\theta_{i,1}$ 
4: for  $t = 2, \dots, T$  do
5:    $z_{i,t} \leftarrow 0$ ,  $\theta_{i,t} \leftarrow \theta_{i,t-1}$ , and  $\tau_{i,t} \leftarrow \tau_{i,t-1}$ 
6:   Receive  $q_{i,t} = \kappa \widehat{C}_i^{\text{KL}}(\theta_t, \theta_{t-1})$  or  $d_t = \|\theta_t - \theta_{t-1}\|_F$ 
7:    $Q_i \leftarrow Q_i + q_{i,t}$  or  $P_i \leftarrow P_i + d_t$ 
8:   if  $Q_i > \Gamma_i$  or  $P_i > \Delta_i$  then
9:      $z_{i,t} \leftarrow 1$ ,  $\theta_{i,t} \leftarrow \theta_t$ , and  $\tau_{i,t} \leftarrow t$ 
10:     $Q_i \leftarrow 0$  or  $P_i \leftarrow 0$ 
11:   end if
12:   Observe  $s_{i,t}$  and select  $a_{i,t}$  using  $\theta_{i,t}$ 
13: end for
```

We next establish the regret-refresh guarantee for the KL-AOINTSYNC instance of Algorithm 1. Let $H := T - 1$ and $W_{i,T}^{\text{KL}} := \sum_{t=2}^T q_{i,t}$ denote the total empirical predictive variation after initialization. Let $\Gamma_i \in \mathbb{R}_+$ be the device-specific cumulative-certificate threshold. Define the horizon-wide event $\mathcal{E}_i := \bigcap_{t=2}^T \{C_{i,t}^{\text{KL}} < q_{i,t}\}$. Under (3.1), the union bound gives $\Pr(\mathcal{E}_i) \geq 1 - \delta$.

On $\mathcal{E}_i$, the recursion above ensures that $Q_i(t)$ controls the stale-model regret accumulated since the latest refresh. Moreover, $Q_i(t) \leq \Gamma_i$ on every non-refresh round. Thus, smaller thresholds provide tighter high-probability control of stale-model regret but generally induce more refreshes. For each accepted $q_{i,\ell}$, let $\rho_{i,\ell}$ count the subsequent non-refresh rounds, including round ℓ , in which it remains in Q_i ; set $\rho_{i,\ell} = 0$ if it triggers a refresh.

THEOREM 3.1 (RESIDENCE-TIME REGRET-REFRESH TRADEOFF). *Let $T \geq 2$, $\Gamma_i > 0$, and fix μ_i , $\mathcal{A}_i$, and h_i over the horizon. Suppose that the probe requirement (3.1) holds for every adjacent model pair. Then, with probability at least $1 - \delta$, KL-AOINTSYNC satisfies*

$$\text{Reg}_i(T) < \sum_{\ell=2}^T q_{i,\ell} \rho_{i,\ell} \leq H\Gamma_i, \quad (7a)$$

$$M_i(T) \leq \max\{0, \lceil W_{i,T}^{\text{KL}} / \Gamma_i \rceil - 1\} \leq W_{i,T}^{\text{KL}} / \Gamma_i, \quad (7b)$$

$$\text{Reg}_i(T) + c_i M_i(T) < H\Gamma_i + c_i W_{i,T}^{\text{KL}} / \Gamma_i. \quad (7c)$$

The residence-time bound may be sharper than $H\Gamma_i$ when changes have short residence times or the accumulated certificate is usually well below the threshold.

The first bound shows that, on the high-probability event $\mathcal{E}_i$, each accepted predictive change contributes in proportion to its residence time in the accumulator. A change followed quickly by synchronization therefore affects few decisions, whereas a persistent change contributes repeatedly. The coarser bound $H\Gamma_i$ follows because the accumulator remains below Γ_i on every non-refresh round. The refresh-count bound, in contrast, follows directly from the deterministic threshold recursion applied to the observed certificates $q_{i,t}$. Each synchronization requires more than Γ_i accumulated empirical predictive variation. Thus, decreasing the threshold provides tighter high-probability control of stale-decision regret but increases refreshes, while increasing it has the opposite effect.

PROOF. Consider the event $\mathcal{E}_i$. By Lemma 3.1, on every non-refresh round, $\text{AoInt}_{i,t} < Q_i(t)$. On a refresh round, θ_t is cached before acting and hence $\text{AoInt}_{i,t} = r_{i,t}(\theta_t) = 0$. Therefore, on $\mathcal{E}_i$, $\text{Reg}_i(T) < \sum_{t:z_{i,t}=0} Q_i(t) = \sum_{\ell=2}^T q_{i,\ell} \rho_{i,\ell}$. The equality counts each accepted certificate once for every non-refresh round in which it remains in the accumulator. Because $Q_i(t) \leq \Gamma_i$ on every such round and there are at most H post-initialization rounds, $\text{Reg}_i(T) < H\Gamma_i$.

For the refresh count, let $t_1 < \dots < t_m$ be the $m = M_i(T)$ refresh rounds and set $t_0 = 1$. A refresh at t_j requires $\sum_{\ell=t_{j-1}+1}^{t_j} q_{i,\ell} > \Gamma_i$. These triggering segments are disjoint. Summing them gives $m\Gamma_i < W_{i,T}^{\text{KL}}$. Since m is a nonnegative integer, $m \leq \max\{0, \lceil W_{i,T}^{\text{KL}} / \Gamma_i \rceil - 1\}$. This refresh-count statement follows directly from the observed certificate sequence and does not require $\mathcal{E}_i$. Combining it with the regret bound on $\mathcal{E}_i$ gives $\text{Reg}_i(T) + c_i M_i(T) < H\Gamma_i + c_i W_{i,T}^{\text{KL}} / \Gamma_i$. Finally, (3.1) gives $\Pr(\mathcal{E}_i) \geq 1 - \delta$, completing the proof. $\square$

COROLLARY 3.1 (ENVELOPE-OPTIMAL FIXED THRESHOLD). *Suppose $0 < W_{i,T}^{\text{KL}} \leq \bar{W}_{i,T}$ is known before operation. Under the conditions of Theorem 3.1, among fixed thresholds, the choice $\Gamma_i^* = \sqrt{c_i \bar{W}_{i,T} / H}$ uniquely minimizes the corresponding worst-case envelope and, with probability at least $1 - \delta$, guarantees*

$$\text{Reg}_i(T) < \sqrt{c_i H \bar{W}_{i,T}}, \quad M_i(T) < \sqrt{H \bar{W}_{i,T} / c_i}, \quad (8a)$$

$$\text{Reg}_i(T) + c_i M_i(T) < 2\sqrt{c_i H \bar{W}_{i,T}}. \quad (8b)$$

For $c_i = \Theta(1)$, $\bar{W}_{i,T} = O(T)$ gives an $O(T)$ cumulative-regret bound, while $\bar{W}_{i,T} = o(T)$ ensures sublinear cumulative regret, and hence vanishing average regret.

This corollary makes the threshold tradeoff explicit. The optimal fixed threshold follows from balancing the high-probability stale-model regret envelope against the synchronization cost.

PROOF. Let $g(\Gamma) := H\Gamma + c_i \bar{W}_{i,T} / \Gamma$ ($\Gamma > 0$). Since $g''(\Gamma) = 2c_i \bar{W}_{i,T} / \Gamma^3 > 0$, g is strictly convex. Solving $g'(\Gamma) = 0$ gives the minimizer $\Gamma_i^* = \sqrt{c_i \bar{W}_{i,T} / H}$. Substituting this value into the bounds of Theorem 3.1 gives the stated results. Moreover, $\text{Reg}_i(T) / T < \sqrt{c_i (H/T) (\bar{W}_{i,T} / T)}$, converging to 0 when $\bar{W}_{i,T} = o(T)$. $\square$

3.2 Parameter-Aware Age-of-Intelligence Synchronization

Although KL-AOINTSYNC directly tracks client-specific predictive changes, evaluating adjacent models on a separate probe set for every device introduces additional inference, communication, and storage overhead that may become substantial in large-scale, resource-constrained IoT systems. To obtain a lighter synchronization mechanism, we therefore relate the KL certificate to parameter changes measured by the fixed Frobenius norm $\|\cdot\|_F$. This relation enables PARAM-AOINTSYNC to replace client-specific probe inference with a single scalar parameter-change signal computed by the server and shared across devices. We first specify the conditions under which the proposed algorithm is applied.

ASSUMPTION 3.1. *Along the model path of interest, there is a finite L_i such that $C_i^{\text{KL}}(\theta, \theta') \leq L_i \|\theta - \theta'\|_F$.*

Assumption 3.1 does not imply that every parameter change affects the device's decisions. Instead, it provides a conservative

sensitivity bound along the relevant model path. Unlike the empirical KL certificate used by KL-AOINTSYNC, this condition directly controls the population certificate and therefore does not introduce finite-probe estimation uncertainty.

The following lemma verifies this condition for a softmax model, where W and W' are two weight matrices in $\mathbb{R}^{|\mathcal{Y}| \times |\mathcal{Y}|}$ and $\|\cdot\|_F$ denotes the Frobenius norm, with $\|W\|_F := \sqrt{\sum_{j=1}^{|\mathcal{Y}|} \sum_{k=1}^{|\mathcal{Y}|} W_{jk}^2}$.

LEMMA 3.2 (SOFTMAX PARAMETER-TO-KL SMOOTHNESS). *Let $p_W(\cdot | s) = \text{softmax}(W^\top \phi(s))$, where $\phi : \mathcal{S} \rightarrow \mathbb{R}^d$ is the feature representation used by the softmax layer. Furthermore, we apply a fixed ℓ_2 normalization before the softmax layer: $\phi(s) = R_i \tilde{\phi}(s) / \max\{\|\tilde{\phi}(s)\|_2, R_i\}$, where $\tilde{\phi}(s)$ denotes the original feature vector. We set R_i to the prescribed feature-norm bound used in this normalization and consequently use $L_i = R_i / (2\sqrt{2})$ in PARAM-AOINTSYNC. Suppose that $\|\phi(s)\|_2 \leq R_i$ for every s in the support of μ_i . Then $C_i^{\text{KL}}(W, W') \leq R_i \|W - W'\|_F / (2\sqrt{2})$. Thus, the smoothness condition holds under the Frobenius norm with $L_i = R_i / (2\sqrt{2})$.*

PROOF. Let $A(z) := \log \sum_y e^{z_y}$ and $[p_z]_y := e^{z_y} / \sum_{y'} e^{z_{y'}}$. Then $\nabla A(z) = p_z$ and $\nabla^2 A(z) = \text{diag}(p_z) - p_z p_z^\top$. For any u , $u^\top \nabla^2 A(z) u = \text{Var}_{Y \sim p_z}(u_Y)$. Popoviciu's variance bound and $\max_y u_y - \min_y u_y \leq \sqrt{2} \|u\|_2$ imply $u^\top \nabla^2 A(z) u \leq (1/2) \|u\|_2^2$; hence, A is $1/2$ -smooth under the Euclidean norm. For a fixed s , let $z = W^\top \phi(s)$ and $z' = W'^\top \phi(s)$. Expressing the categorical KL divergence as the corresponding Bregman divergence of A gives $D_{\text{KL}}(p_W(\cdot | s) \| p_{W'}(\cdot | s)) \leq (1/4) \|z - z'\|_2^2 \leq R_i^2 \|W - W'\|_F^2 / 4$. Taking expectation over $s \sim \mu_i$, multiplying by $1/2$, and taking the square root yields $C_i^{\text{KL}}(W, W') \leq R_i \|W - W'\|_F / (2\sqrt{2})$.

In the experiments, we enforce the required feature bound using the fixed normalization $\phi(s) = R_i \tilde{\phi}(s) / \max\{\|\tilde{\phi}(s)\|_2, R_i\}$, where $\tilde{\phi}(s)$ is the original feature vector. Therefore, $\|\phi(s)\|_2 \leq R_i$, with given R_i and $L_i = R_i / (2\sqrt{2})$. $\square$

PROPOSITION 3.1 (PARAMETER VARIATION CONTROLS AOINT AND REGRET). *Let $d_t := \|\theta_t - \theta_{t-1}\|_F$ and $P_i(t) := \sum_{\tau=t+1}^T d_\tau$. Under Assumption 3.1, $\text{AoInt}_{i,t} = r_{i,t}(\theta_{i,t}) \leq L_i P_i(t)$.*

$P_i(t)$ is the parameter-space path length accumulated while device i retains its cached model. Because successive parameter changes may cancel or occur in directions that barely affect model outputs, this certificate can be conservative. Nevertheless, under Assumption 3.1, it provides a deterministic upper bound on AoInt and stale-model regret and does not require finite-probe estimation.

PROOF. Fix t and let $\tau := \tau_{i,t}$. By the same path decomposition used in Lemma 3.1, $\text{AoInt}_{i,t} \leq \sum_{\ell=\tau+1}^t C_i^{\text{KL}}(\theta_\ell, \theta_{\ell-1})$. Applying Assumption 3.1 to each adjacent pair gives $\text{AoInt}_{i,t} \leq L_i \sum_{\ell=\tau+1}^t \|\theta_\ell - \theta_{\ell-1}\|_F = L_i P_i(t)$, which proves the result. $\square$

We can now obtain the analogous regret-refresh tradeoff for the PARAM-AOINTSYNC instance of Algorithm 1 by replacing predictive variation with accumulated parameter variation. Unlike the KL result, the following bound is deterministic under Assumption 3.1 and therefore does not require the high-probability event $\mathcal{E}_i$.

COROLLARY 3.2 (PARAM-AOINTSYNC REGRET-REFRESH TRADEOFF). *Let $V_T^\theta := \sum_{t=2}^T d_t$, and let $\sigma_{i,\ell}$ count the non-refresh rounds for*

Table 1: Performance comparison on 20 fresh validation traces. Values are mean $\pm$ standard deviation.

Method	Regret	Refreshes	Total Cost
Param-AoIntSync	430.32 $\pm$ 82.66	84.40 $\pm$ 5.49	451.42 $\pm$ 83.08
KL-AoIntSync	432.53 $\pm$ 80.71	74.20 $\pm$ 2.67	451.08 $\pm$ 80.99
Instant-Param	474.02 $\pm$ 135.03	28.00 $\pm$ 11.16	481.02 $\pm$ 134.19

which an accepted d_ℓ remains in P_i . Define $S_i := \sum_{\ell=2}^T d_\ell \sigma_{i,\ell}$. Then the PARAM-AOINTSYNC instance of Algorithm 1 satisfies

$$\text{Reg}_i(T) \leq L_i S_i \leq L_i H \Delta_i, \quad M_i(T) \leq V_T^\theta / \Delta_i, \quad (9a)$$

$$\text{Reg}_i(T) + c_i M_i(T) \leq L_i H \Delta_i + c_i V_T^\theta / \Delta_i. \quad (9b)$$

If $V_T^\theta \leq \bar{V}_T^\theta$, the last envelope is minimized by $\Delta_i^* = \sqrt{c_i \bar{V}_T^\theta / (L_i H)}$, and is at most $2\sqrt{L_i c_i H \bar{V}_T^\theta}$. Hence, constant c_i, L_i and $\bar{V}_T^\theta = o(T)$ imply sublinear cumulative stale-model regret.

This result mirrors the KL-AOINTSYNC tradeoff, with the parameter-path budget V_T^θ replacing predictive variation. The sensitivity constant L_i maps the server model's common parameter motion to a client-specific deterministic bound on output variation.

PROOF. Let $\mathcal{N}_i$ be the non-refresh rounds after initialization. For $t \in \mathcal{N}_i$, Proposition 3.1 gives $\text{AoInt}_{i,t} = r_{i,t}(\theta_{i,t}) \leq L_i P_i(t)$, whereas refresh rounds have zero stale-model regret. Summing over $\mathcal{N}_i$ and exchanging the two finite sums yields $\text{Reg}_i(T) \leq L_i \sum_{t \in \mathcal{N}_i} P_i(t) = L_i \sum_{\ell=2}^T d_\ell \sigma_{i,\ell}$. The equality counts each accepted increment d_ℓ once for every non-refresh round in which it remains in P_i . Since $P_i(t) \leq \Delta_i$ and $|\mathcal{N}_i| \leq H$, $\text{Reg}_i(T) \leq L_i H \Delta_i$.

For the refresh count, write the refresh rounds as $t_1 < \dots < t_m$ and set $t_0 = 1$. Each triggering segment contains more than Δ_i path length, and the segments are disjoint. Hence, $m \Delta_i < V_T^\theta$, which gives $M_i(T) \leq V_T^\theta / \Delta_i$. Adding $c_i M_i(T)$ gives $\text{Reg}_i(T) + c_i M_i(T) \leq L_i H \Delta_i + c_i V_T^\theta / \Delta_i$. With $V_T^\theta \leq \bar{V}_T^\theta$, define $f(\Delta) = L_i H \Delta + c_i \bar{V}_T^\theta / \Delta$. Since $f''(\Delta) = 2c_i \bar{V}_T^\theta / \Delta^3 > 0$, f is strictly convex. Solving $f'(\Delta) = 0$ gives $\Delta_i^* = \sqrt{c_i \bar{V}_T^\theta / (L_i H)}$. Substitution gives $\text{Reg}_i(T) + c_i M_i(T) \leq 2\sqrt{L_i c_i H \bar{V}_T^\theta}$. Finally, $\text{Reg}_i(T)/T \leq \sqrt{L_i c_i (H/T) (\bar{V}_T^\theta / T)} \rightarrow 0$, $\bar{V}_T^\theta = o(T)$. $\square$

4 Numerical Results

We evaluate whether the proposed methods realize the predicted regret-refresh tradeoff and whether cumulative model changes improve synchronization. We compare Param-AoIntSync and KL-AoIntSync with *Instant-Param*, which refreshes when $\|\theta_t - \theta_{t-1}\|_F > \delta$ without accumulating sub-threshold changes.

We consider an exogenous three-class softmax model over $T = 600$ communication rounds and $N = 8$ heterogeneous clients. The server model evolves through gradual Gaussian parameter drift with standard deviation 0.006 together with six abrupt changes of Frobenius norm 1.6. Each client has a finite population of 1024 contexts and 96 fixed unlabeled probes. The refresh cost is $c_i = 0.25$. The synchronization thresholds are selected using calibration traces and then frozen before evaluation on 20 fresh validation traces.

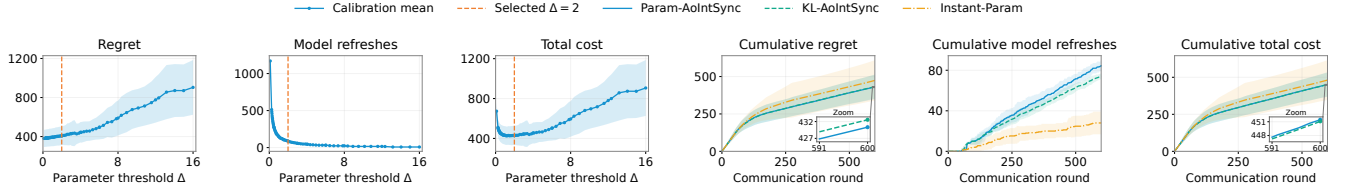


Figure 3: Regret and refresh-cost tradeoff. The first three panels show Param-AoIntSync’s calibration results as its threshold varies; the vertical dashed line marks the selected threshold $\Delta = 2$. The last three panels compare cumulative regret, refresh count, and total cost over 20 fresh validation traces. Main-panel shading denotes one sample standard deviation.

We first compare the overall performance of the proposed cumulative synchronization policies with the non-accumulating Instant-Param baseline. Table 1 reports the mean and standard deviation of regret, number of model refreshes, and total cost over 20 fresh validation traces. As shown in Table 1, Param-AoIntSync and KL-AoIntSync achieve mean regrets of 430.32 and 432.53, respectively, compared with 474.02 for Instant-Param. Their corresponding total costs are 451.42 and 451.08, both lower than the 481.02 obtained by Instant-Param. This improvement is achieved at the expense of more frequent synchronization: Param-AoIntSync and KL-AoIntSync perform 84.40 and 74.20 refreshes on average, respectively, whereas Instant-Param performs only 28.00.

To further illustrate the regret–refresh tradeoff, we examine how cumulative regret and refresh cost vary with the synchronization threshold and evolve over time in Fig. 3. Figure 3 first illustrates the calibration of Param-AoIntSync under the true-action regret metric. Increasing the parameter threshold Δ generally reduces the number of model refreshes by allowing the cached model to tolerate larger accumulated changes before synchronization. This reduction in communication comes at the cost of increased decision loss, producing the expected regret–communication tradeoff. Under this objective, the selected Param-AoIntSync threshold remains $\Delta = 2$, while the KL-AoIntSync threshold is adjusted from 0.2 to 0.3152.

The regret trajectories in Fig. 3 make the resulting performance difference visible. Although Instant-Param communicates substantially less frequently, its cumulative decision loss grows above those of Param-AoIntSync and KL-AoIntSync. The additional refreshes help prevent decision loss that would otherwise accumulate under the non-accumulating baseline. Overall, by accumulating small model changes, our methods achieve lower true-action regret and total cost than Instant-Param, at the expense of more refreshes.

5 Conclusion

We introduced AoInt, a decision-aware measure of model freshness, and developed KL-AoIntSync and the lightweight PARAM-AoIntSync with regret–refresh guarantees. Experiments show that both methods achieve lower observed mean true-action regret and total cost than a non-accumulating parameter-change trigger, at the expense of more refreshes. Neither requires online labels or rewards, while PARAM-AoIntSync avoids client-specific probe evaluation.

Future work can entail extending our framework to simultaneous model and environment drift, joint training and deployment, integration of application-centric objectives and trade-offs, and co-ordinated multi-device synchronization. Also, it can entail exploring continual federated learning checkpoints, certificate conservatism

for larger models, adaptive distance measures and threshold calibration under changing model dynamics, and deployment in emerging settings such as spatial intelligence under smart AI glasses.

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